

METAL SOLUTIONS

EOS StainlessSteel 17-4PH

Material Data Sheet

EOS STAINLESSSTEEL 17-4PH

EOS StainlessSteel 17-4PH is a martensitic precipitation hardening stainless steel with high strength and toughness. EOS 17-4PH corresponds to ASTM A564 (UNS17400, 17-4PH). 17-4PH can be heat treated for use in components with different requirements. It can be widely used in engineering solutions and in components in the medical field.

MAIN CHARACTERISTICS

- High strength
- Good corrosion resistance

TYPICAL APPLICATIONS

- Surgical tools
- Pump components
- Paper mill equipment

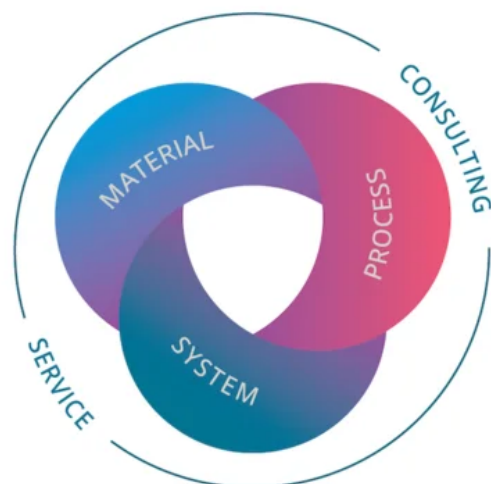
The EOS Quality Triangle

EOS uses an approach that is unique in the AM industry, taking each of the three central technical elements of the production process into account: the system, the material and the process. The data resulting from each combination is assigned a Technology Readiness Level (TRL) which makes the expected performance and production capability of the solution transparent.

EOS incorporates these TRLs into the following two categories:

- Premium products (TRL 7-9): offer highly validated data, proven capability and reproducible part properties.
- Core products (TRL 3 and 5): enable early customer access to newest technology still under development and are therefore less mature with less data.

All of the data stated in this material data sheet is produced according to EOS Quality Management System and international standards



POWDER PROPERTIES

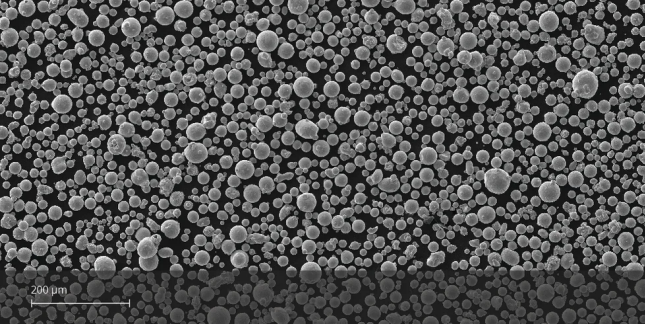
The chemical composition of EOS stainlessSteel 17-4PH is in compliance with ASTM A564.

Powder Chemical Composition (wt.-%)

Element	Min.	Max.
Cr	15	17.5
Ni	3	5
Cu	3	5
Si	0	1
Mn	0	1
C	0	0.07
P	0	0.04
S	0	0.03
Nb + Ta	0.15	0.45

Powder Particle Size

Generic Particle Size Distribution	15 - 65 µm
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HEAT TREATMENT

Description

EOS 17-4PH can be heat treated to match various needs of different applications. The two step heat treatment can be performed under vacuum or inert gas atmosphere. First step is solution annealing to minimize amount of austenite in the martensitic matrix. The needed hardness and strength is achieved through aging treatment where hardening takes place through forming of precipitates.

Steps

Solution annealing:
0.5-1 h at 1040 °C (±15 °C) measured from the part followed by rapid air cooling to room temperature (below 32 °C). Cooling rate 5-60 °C/min. Reaching room temperature before starting aging treatment is required to achieve desired microstructure.

Ageing:
Peak hardness of 42 HRC is achieved with H900 ageing treatment of 1 h at 480 °C (±10 °C) measured from the part followed by air cooling. Mechanical properties presented in this document achieved through this aging procedure. For lower hardness and strength choose aging procedure according to the table.

Condition	Ageing temperature [°C]	Hold time [h]	Cooling	Hardness [HRC]
H900	480	1	Air	42
H925	495	4	Air	38
H1025	550	4	Air	36
H1075	580	4	Air	34
H1100	595	4	Air	32
H1150	620	4	Air	28

CYTOTOXICITY

Cytotoxicity tests were done according to ISO 10993-5. It included growth inhibition tests evaluated from two endpoints (XTT & BCA). Tests were done with as-manufactured cubes. EOS StainlessSteel 17-4PH cubes were extracted under agitation for 24±2h with DMEM 10% FBS. L929 cells were then incubated for 68 – 72h with the following concentrations of the test extract: 13.2%, 19.8%, 29.6%, 44.4%, 66.7% and 100%. Surface/volume ratio used was 3cm²/mL.

In this study under the given conditions no leachable substances were released in cytotoxic concentrations from the test item as confirmed by two different endpoints (XTT, BCA).

EOS StainlessSteel 17-4PH for EOS M 290 | 40/80 µm

EOS M 290 - 40 µm - TRL 3

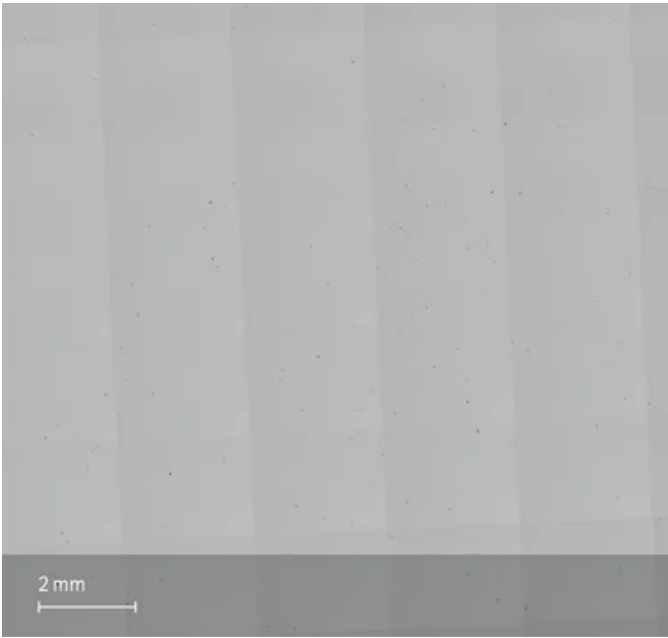
Process Information Metal

This process product is optimized for flexible and fast production of 17-4PH parts with EOS M 290 system. The parameter set has three different layer thickness options that can all be utilized within the same build: 40µm, 80 µm and 40/80 µm Skin. For high productivity needs a 80 µm parameter set is included with a build rate more than double the long established 40 µm parameter set. Both can be used separately for different parts or combined by using 40/80 µm Skin with faster 80 µm for the core of the part and higher quality 40 µm building for the surface of the part with defined thickness. Sectionioning parts in vertical direction for different parameters is also possible.

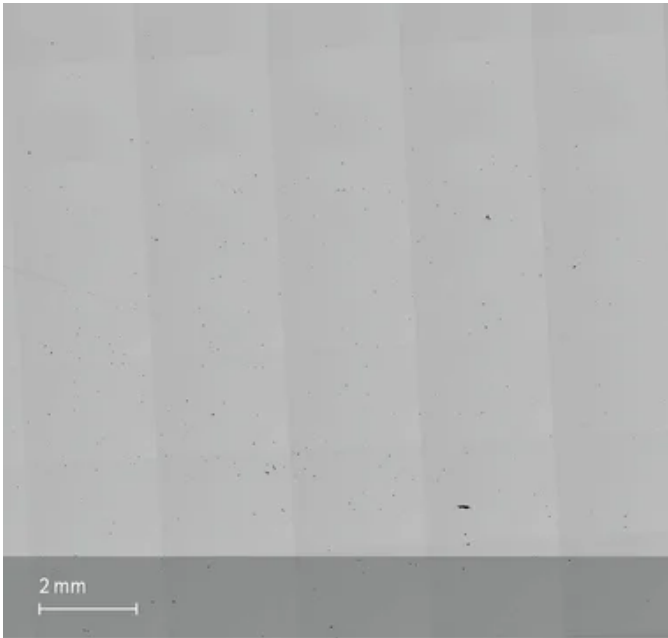
System Setup	EOS M 290
EOS Material set	17-4PH_040_080_Core M291
Required Software Plan	EOS BUILD
Software Requirements	EOSPRINT 2.14 or newer EOSYSTEM 2.18 or newer
Recoater Blade	Ceramic
Nozzle	EOS Grid Nozzle
Inert gas	Argon
Sieve	75 µm

Additional Information	
Layer Thickness	40 µm, 80 µm, 40/80 µm
Volume Rate	4.1-7.9 mm³/s
Typical Dimensional Change after HT [%]	+0.2 %

Chemical and Physical Properties of Parts



As manufactured, 40 μm



As manufactured, 80 μm

Etched micrograph of powder

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	40 μm	<0.1 %	-
Average Defect Percentage	80 μm	<0.2 %	-

Mechanical Properties

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 40 μm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1240	1360	14	-	-	-
Horizontal	1240	1360	14	-	-	-

H900 Heat treatment

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 80 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1240	1360	13	-	-	-
Horizontal	1230	1355	14	-	-	-

H900 Heat treatment

Impact Toughness

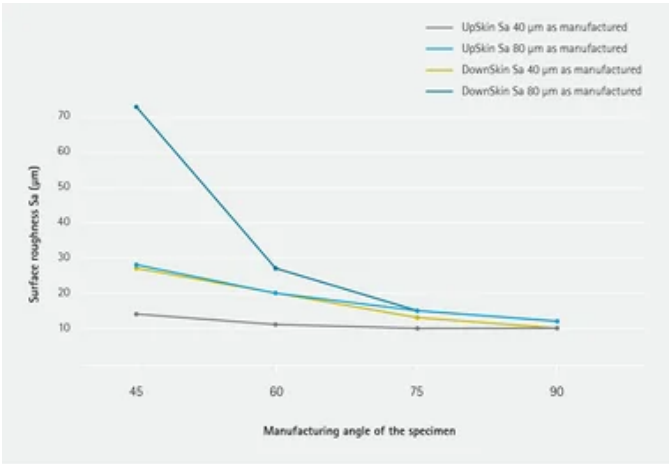
EN ISO 148-1, Charpy-V Heat Treated	
Typical Impact Toughness [J]	20
Layer Thickness	40 µm

H900 Heat treatment

Heat Treated	
Typical Impact Toughness [J]	20
Layer Thickness	80 µm

H900 Heat treatment

Surface Roughness



Coefficient of Thermal Expansion

ASTM E228	Temperature
10.4*10 ⁻⁶ /K	25 – 100 °C
11.1*10 ⁻⁶ /K	25 – 200 °C
11.4*10 ⁻⁶ /K	25 – 300 °C
11.8*10 ⁻⁶ /K	25 – 400 °C

EOS StainlessSteel 17-4PH on EOS M 400-4 | 40/80 µm

EOS M 400-4 - 40 µm - TRL 3

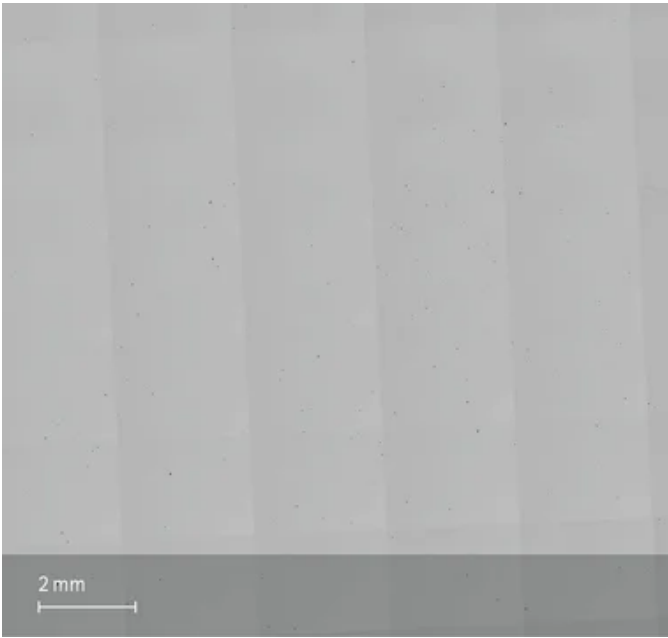
Process Information Metal

This process product is optimized for flexible and fast production of 17-4PH parts with M 400-4 system. The parameter set has three different layer thickness options that can all be utilized within the same build: 40µm, 80 µm and 40/80 µm Skin. For high productivity needs a 80 µm parameter set is included with a build rate more than double the long established 40 µm parameter set. Both can be used separately for different parts or combined by using 40/80 µm Skin with faster 80 µm for the core of the part and higher quality 40 µm building for the surface of the part with defined thickness. Sectionioning parts in vertical direction for different parameters is also possible.

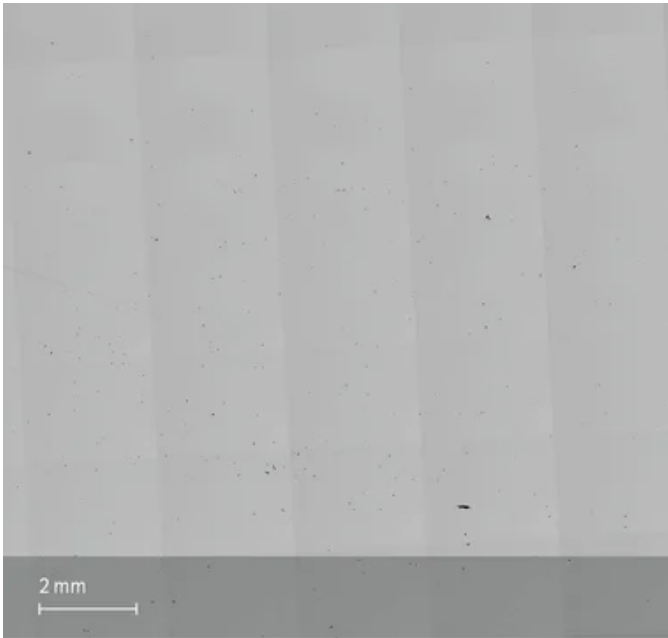
System Setup	EOS M 400-4
EOS Material set	17-4PH_040_080_Core M404
Required Software Plan	EOS BUILD
Software Requirements	EOSPRINT 2.14 or newer EOSYSTEM 2.18 or newer
Recoater Blade	Ceramic
Inert gas	Argon
Sieve	75 µm

Additional Information	
Layer Thickness	40 µm, 80 µm, 40/80 µm
Volume Rate	4.1 -7.9 mm³/s
Typical Dimensional Change after HT [%]	+0.2 %

Chemical and Physical Properties of Parts



As manufactured, 40 μm



As manufactured, 80 μm

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	40 μm	<0.1 %	-
Average Defect Percentage	80 μm	<0.2 %	-

Mechanical Properties

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 40 μm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1245	1350	13	-	-	-
Horizontal	1240	1360	13	-	-	-

H900 Heat treatment

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 80 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1190	1350	14	-	-	-
Horizontal	1180	1360	14	-	-	-

H900 Heat treatment

Impact Toughness

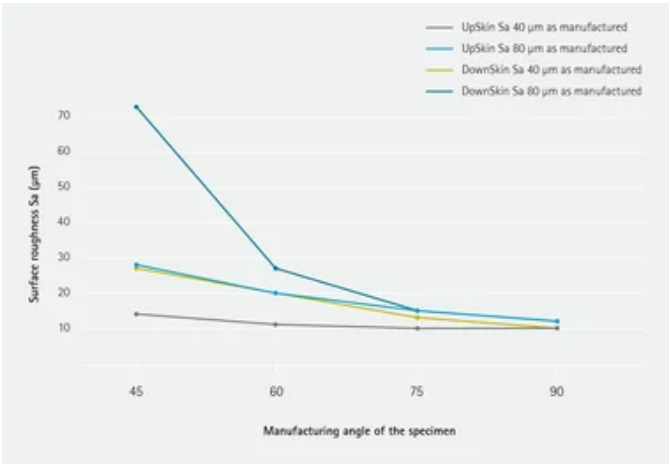
EN ISO 148-1, Charpy-V Heat Treated	
Typical Impact Toughness [J]	20
Layer Thickness	40 µm

H900 Heat treatment

Heat Treated	
Typical Impact Toughness [J]	20
Layer Thickness	80 µm

H900 Heat treatment

Surface Roughness



Coefficient of Thermal Expansion

ASTM E228	Temperature
10.4*10 ⁻⁶ /K	25 – 100 °C
11.1*10 ⁻⁶ /K	25 – 200 °C
11.4*10 ⁻⁶ /K	25 – 300 °C
11.8*10 ⁻⁶ /K	25 – 400 °C

EOS StainlessSteel 17-4PH for EOS M 290 | 40 µm

EOS M 290 - 40 µm - TRL 7

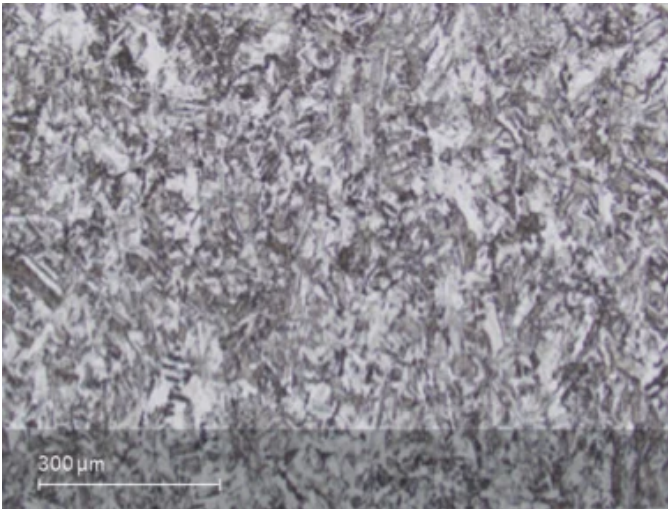
Process Information Metal

This process product is optimized for flexible and reliable production of 17-4PH parts with emphasis on quality.

System Setup	EOS M 290
EOS Material set	17-4PH_040_StainlessM291
Software Requirements	EOSPRINT 1.5 or newer EOSYSTEM 2.5 or newer
Recoater Blade	Ceramic
Nozzle	EOS Grid Nozzle
Inert gas	Argon
Sieve	75 µm

Additional Information	
Layer Thickness	40 µm
Volume Rate	3.3 mm³/s
Typical Dimensional Change after HT [%]	+0.2 %

Chemical and Physical Properties of Parts



Heat treated. Etchant: Marble

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
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Average Defect Percentage	-	0.03 %	-
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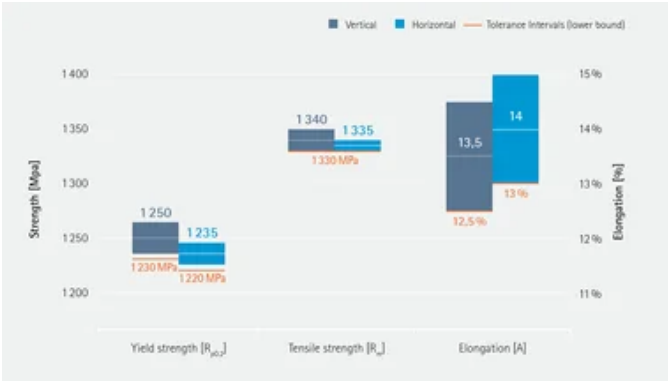
Density EN ISO 3369	Thickness	Result	Number of Samples
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Average Density	-	≥7.79 g/cm³	-
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Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 40 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1250	1340	13.5	-	-	-
Horizontal	1235	1335	14	-	-	-

H900 Heat treatment



T95: One-sided (lower bound) Tolerance Interval provides limit above which 9x% of the population falls with 95% level of confidence. Tolerance intervals are based on e.g validation data / QA statistics.

Mechanical Properties As Manufactured

EN ISO 6892-1 Room Temperature	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	860	920	20	-	-	-
Horizontal	890	860	20	-	-	-

Hardness

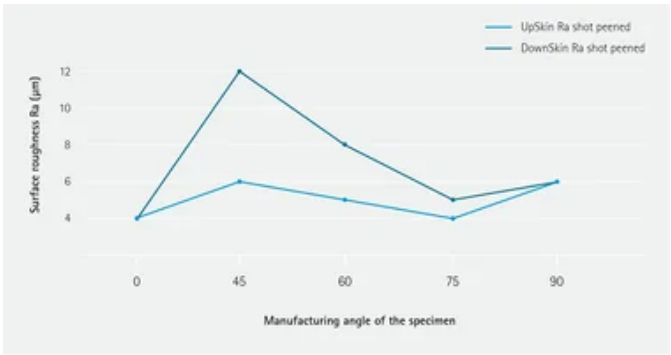
EN ISO 6508 Heat Treated	
Value	42
Unit	HRC

EN ISO 6508 As Manufactured

Value 24

Unit HRC

Surface Roughness



Coefficient of Thermal Expansion

ASTM E228	Temperature
10.4*10 ⁻⁶ /K	25 – 100 °C
11.1*10 ⁻⁶ /K	25 – 200 °C
11.4*10 ⁻⁶ /K	25 – 300 °C
11.8*10 ⁻⁶ /K	25 – 400 °C

EOS StainlessSteel 17-4PH for EOS M 300-4 | 40 µm

EOS M 300-4 - 40 µm - TRL 3

System Setup	EOS M 300-4
EOS Material set	17-4PH_040_CoreM304
Software Requirements	EOSPRINT 2.11 or newer EOSYSTEM 2.15 or newer
Recoater Blade	Ceramic
Build Platform Temperature	80 °C
Inert gas	Argon
Sieve	75 µm

Additional Information	
Layer Thickness	40 µm
Volume Rate	up to 4 x 3.32 mm³/s

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	-	0.014 %	-

Mechanical Properties

Mechanical Properties As Manufactured

EN ISO 6892-1 Room Temperature	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	947	1052	15.7	-	-	-
Horizontal	941	1015	16.8	-	-	-

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1240	1371	10.5	-	-	-
Horizontal	1207	1349	10.7	-	-	-

HEADQUARTERS

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